

EGE UMUT CAN

Intended Department: Economics, Finance & Quantitative Data Science

Istanbul, Türkiye | FMV Isik Schools, Ispartakule Campus (Class of 2027) | **Live Data Lab:** ege211.github.io/macroeconomic-data-lab |

GitHub: github.com/ege211

EDUCATION & QUALIFICATIONS

FMV Özel Ispartakule Isik High School (FMV Isik Schools), Istanbul, Türkiye

Sept 2023 – June 2027

- **Academic Standing:** Cumulative GPA: **90.0 / 100.0** (11th Grade, High Honour / Honour Roll Standing)
- **Advanced Placement (AP) Coursework:** AP Macroeconomics, AP Microeconomics, AP Calculus, AP Psychology
- **Core Research Areas:** Monetary Policy Credibility & Inflation Persistence, Capital Market Microstructure, Financial Machine Learning, Strategic Capital Allocation

SUMMER SCHOOLS, INTERNATIONAL CERTIFICATIONS & INSTITUTIONAL SIMULATIONS

Docklands Academy London — Intensive English & UK Academic Summer Program, London, UK

2024 – 2025 (20 hrs/wk)

Student; Accredited by the British Council & British Accreditation Council (BAC), Member of English UK & Study UK

- **Progressive CEFR Certifications (B2, B2+ & C1 Advanced):** Completed intensive residential and academic English modules in London (including the June 10–21, 2024 London session), earning official progression certificates across **CEFR B2 (Upper-Intermediate)**, **B2+**, and **C1 (Advanced Proficient User)** levels.
- **Academic & Cultural Immersion:** Engaged in seminar discussions, analytical writing, and cross-cultural communication in London's Canary Wharf / Docklands financial district, while conducting observational field research on UK hospitality service-profit chains.

Bank of America — Global Markets Sales & Trading & Private Bank Simulations (Forage)

June – Sept 2026 (6 hrs/wk)

Analyst Participant; Verified Completion IDs: 8bD49n4kfWMyweRGZ (Global Markets) & 6ab97c46a90bc149453c6cc5 (Private Bank)

- **Global Markets Sales & Trading:** Authored a *Market Insight Report*, a *Strategic Recommendation Proposal* (Summit Investments), and a *Process Improvement Proposal* (Apex Global Investments) covering live market dynamics and trading-desk workflow automation.
- **The Private Bank Wealth Management:** Formulated multi-generational client asset-allocation frameworks, matching long-term risk tolerance and estate priorities to strategic investment solutions.

Goldman Sachs — Risk Analysis Job Simulation (Forage)

June 2026 (4 hrs/wk)

Analyst Participant; Verified Completion ID: eDZtiYexR4uGgD9Br

- **Quantitative Credit & Real Estate Risk Profiling:** Evaluated institutional client balance-sheet profiles and commercial real-estate investment exposures under downside macroeconomic stress scenarios.

WORK EXPERIENCE & CORPORATE INTERNSHIPS

TEB Arval (BNP Paribas Group), Gayrettepe HQ, Istanbul — Fleet Sales, Operations & HR Intern

Jan 19 – Jan 26, 2026 (40 hrs/wk)

11th Grade Corporate Intern; Supervised by Human Resources & Fleet Sales Organization Department (Reference Letter Awarded)

- **Multinational Corporate Operations:** Completed an intensive full-time corporate internship at TEB Arval (BNP Paribas joint venture), observing B2B operational fleet leasing workflows, organizational process architecture, and corporate client relationship management.
- **Administrative & HR Process Analysis:** Collaborated with the Human Resources and Fleet Sales teams to analyze workflow bottlenecks, contract lifecycle coordination, and cross-departmental KPI tracking across corporate accounts.

Haber Tekstil Dis Ticaret (Bohoo Textile Export Supplier), Istanbul — Export Department Intern

Jan – Feb 2026 (35 hrs/wk)

11th Grade Foreign Trade Intern & Field Researcher; Export Department (Official Completion Letter Awarded)

- **B2B International Export Workflows:** Analyzed international apparel export documentation, customs compliance, and supply-chain scheduling for European retail clients (including UK fast-fashion supply chains).
- **Competitive Value Proposition (CVP) Study:** Documented how relational trust, rapid sample-to-production lead times, and quality assurance insulate mid-market Turkish exporters from pure price competition.

FOUNDER & OPEN-SOURCE MACROECONOMIC PLATFORM

Comparative Macroeconomic Data Lab (GEA) — Founder, Developer & Lead Researcher

10th–11th Grade, 8 hrs/wk,
2025–Present

Live Platform: ege211.github.io/macroeconomic-data-lab | Open-Source Repository: github.com/ege211/macroeconomic-data-lab

- **Live World Bank API Architecture:** Engineered an interactive web laboratory retrieving live **World Bank Open Data API v2** feeds (NY.GDP.MKTP.KD.ZG, FP.CPI.TOTL.ZG) across 5 economies (Türkiye, USA, Germany, UK, Japan; 2019–2023) with CSV data-cube export.
- **Dual Econometric Simulators:** Built two browser-based quantitative simulators in JavaScript/Chart.js: (1) a *2024–2025 Cross-Country Scenario Simulator* modeling policy-rate (+0 to +500 bps) and supply-price (+0% to +20%) shocks; and (2) an *AR(1) Inflation Persistence & Shock Half-Life Simulator* operationalizing Working Paper WP-01 (beta = 0.72, half-life $t_{1/2}$ = 2.11 years vs. 0.58 years under anchored expectations).
- **10-Paper Research Corpus & Intellectual Map:** Authored and published 10 full-text working papers, essays, and Data Lab methodological notes organized across four analytical pillars, supported by a 6-work theoretical reading list (Acemoglu & Robinson, Gali & Gertler, Taylor, Piketty, SPK III-52.1, Porter).

INDEPENDENT ECONOMETRIC & QUANTITATIVE RESEARCH PROJECTS

A Machine-Learning-Based Early Warning Framework for Systemic Banking Crises

11th Grade, 10 hrs/wk, 12 wks (2026)

- **Panel Architecture:** Evaluated macro-financial indicators across a balanced panel of **76 World Bank economies (1990–2025; 684 out-of-sample evaluations)** using penalized Logistic Regression and Random Forest models.
- **Leak-Free Temporal Validation:** Built an expanding-window validation pipeline (T = 2000–2008), isolating train-only preprocessing and cost-sensitive class weighting to address a **2.78% rare-event crisis base rate**.
- **Empirical Diagnostic:** Showed that adding balance-sheet leverage (private credit-to-GDP) and external deficits more than doubled pooled recall to **47.37%** (vs. 21.05% baseline), while diagnosing an out-of-sample ranking breakdown (**ROC-AUC = 0.2301**) outside the 2007 pre-GFC cluster.

AI-Assisted Equity Valuation: A Walk-Forward Study on SEC Form 10-K Filings

11th Grade, 10 hrs/wk, 10 wks (2026)

- **Point-in-Time Data Engineering:** Extracted 12 pre-specified linguistic dimensions across **30 non-financial S&P large-cap firms (2014–2024)** via an automated five-tier SEC EDGAR XBRL normalization cascade.
- **Structural DCF & Null-Result Proof:** Built a deterministic DCF and Reverse DCF engine linked to an auditable qualitative translation bridge (*valuation_bridge_v1.0*); proved out-of-sample that 10-K text signals yield **zero statistically significant incremental power** over trailing fundamentals in forward margin forecasting ($p = 0.2335$), with unconstrained high-dimensional text inducing **-13.14% degradation**.

Strategic Capital Allocation in Investment Banking: G-SIBs vs. Advisory Boutiques

11th Grade, 8 hrs/wk, 10 wks (2026)

- **Primary Archival Dataset:** Hand-collected **540 audited observations across 40 SEC Form 10-K filings (2020–2024)** comparing Global Systemically Important Banks (Panel A) and Independent Advisory Boutiques (Panel B).
- **Key Findings:** Disclosed **\$233.2B in shareholder distributions** (\$113.8B dividends, \$119.3B repurchases) and identified diverging lagged dynamics: near-zero correlation with ROE in BHCs ($r = -0.049$) vs. negative correlation with operating margins in boutiques ($r = -0.403$) driven by Basel III **CET1 buffers (13.1%–17.4%)** vs. partner compensation liquidity.

Inflation Persistence & Monetary Policy Effectiveness: Cross-Country Panel Evidence (WP-01)

11th Grade, 6 hrs/wk, 8 wks (2026)

- **Econometric Specification:** Estimated dynamic panel regressions across **45 advanced and emerging economies (2000–2023)** using World Bank, IMF IFS, and BIS policy-rate data; identified high emerging-market inflation persistence (**beta = 0.72**) and a 2–4 quarter policy transmission lag ($\gamma < 0$).

Sports Finance & Tactical Analytics: Arsenal FC Longitudinal Study (2015/16–2024/25)

11th Grade, 6 hrs/wk, 8 wks (2026)

- **Multi-Source Integration:** Merged 10 seasons of audited corporate accounts (*Arsenal Holdings Limited*), a **128-transaction transfer census**, and *Understat xG/xPTS* metrics; proved transfer spending correlates with tactical process (**xGD $r = +0.5982$; xPTS $r = +0.6189$**) rather than short-run noisy match points (PPG $r = +0.3911$), while challenging single-club wage-bill dominance ($r = +0.0516$).

MACROECONOMIC POLICY, MARKET MICROSTRUCTURE & STRATEGY ESSAYS

- **Borsa Istanbul — The Fragility of Liquidity (2026):** Analyzed Turkish capital market microstructure across the 2022–2023 retail equity boom and May 2026 liquidity squeeze; evaluated mutual fund redemption spirals under **SPK Communiqué III-52.1** and corporate bond (*tahvil*) risk pricing.
- **Exchange Rate Pass-Through (ERPT) in Emerging Markets (2026):** Contrasted short-run import-price pass-through elasticities in emerging markets (**0.38**) vs. advanced economies (**0.09**), examining non-linear depreciation threshold effects.
- **The Credibility Gap & The Lira Doesn't Lie (2026):** Investigated central bank independence, time-inconsistency, behavioral anchoring breakdown, and household dollarization in Türkiye.
- **The Paradox of Petrowealth, Contested Terrain & Inditex Strategy Case Study (2026):** Comparative political-economy essays on Dutch disease (Venezuela vs. Norway/GCC) and Piketty's $r > g$, alongside a Porter's Five Forces / VRIO operational analysis of Inditex's 2–3 week supply chain.

TECHNICAL SKILLS, LANGUAGES & PERSONAL DETAILS

- **Quantitative & Programming:** Python (*pandas*, *scikit-learn*, *numpy*, *statsmodels*, *pytest*), SQL / DuckDB, Panel & Time-Series Econometrics, Walk-Forward Backtesting, Out-of-Sample Validation.
- **Financial & Web Engineering:** SEC EDGAR XBRL Parsing, Deterministic & Reverse DCF Modeling, World Bank REST API v2 Integration, Git / GitHub, HTML5 / CSS3 / JavaScript (*Chart.js*).
- **Languages:** Turkish (Native), English (**CEFR C1 Advanced** — Certified by Docklands Academy London, British Council Accredited; B2, B2+ & C1 Progression Certificates).
- **Personal Details:** Date & Place of Birth: January 17, 2008 · Istanbul, Türkiye | Nationality: Turkish.